Lecture

10/22/26

A directed graph is a DAG iff it has a topological ordering of its vertices

Recall the example from last class:
1, 3, 5, 4, 2

Shortest Paths in DAGs via topological sorting
$G = (V, E)$ $|V| = m$
$(u,v) \in E$ weight (u,v)
L = topological sort of vertices in V
$L[1], L[2], \ldots, L[m]$

Shortest Path Algorithm:
d = array of length m: $d[i]$ = shortest path cost from vertex to i

$d[1] = 0$, $d[u] = \infty$, $2 \leq u \leq m$
P: an array of backtracking pointers
$p[v] = -\infty$, $1 \leq v \leq m$
$p[v]$ = predecessor of vertex v in the shortest path from 1 to v

for $i = 1$ to m do
  $u = L[i]$
  for vertex v such that $(u,v) \in E$:
    let $w = \text{weight}(u,v)$
    if $d[v] > d[u] + w$
      then $d[v] = d[u] + w$
        $p(v) = u$

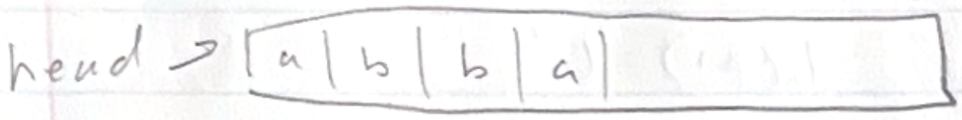
Deterministic Finite Automata PPT!

- Transition graph

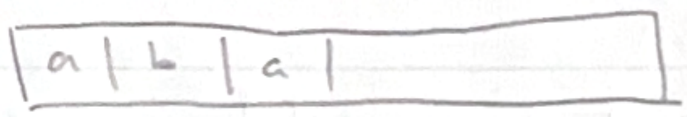
  Alphabet $\Sigma = \{a, b\}$

  Initial state → accepting state

  A certain order of transitions leads us to the accepting state



Rejection case:



q_5 is not the accepting state, so we reject

Another rejection case: empty tape (λ)

- Another example

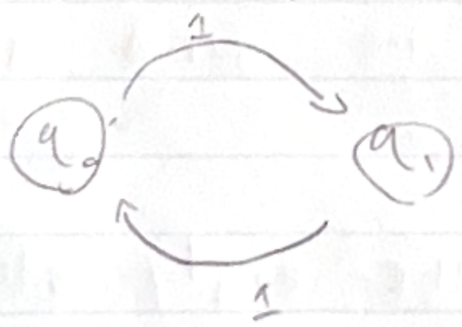
  $L(\lambda, ab, abba)$ we can have multiple accepting states!

- Another example: trap state

  - We can have an infinite set of strings from initial → accepting

$$L = \{a^n b : n \geq 0\}$$

- Another example



EVEN

- Formal definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q: set of states

Σ: input alphabet

δ: transition function

q_0: initial state

F: set of accepting states

- Transition function $\delta: Q \times \Sigma \rightarrow Q$

  $\delta(q, x) = q'$

  ex)

  $\delta(q_0, a) = q_1$

- Extended transition function

  $\delta^*(q, w) = q'$ where w is multiple transitions

- Language accepted by M or A

  We say L' is accepted if $L(M) = L'$

  $L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \in F\}$

- Further examples

  $L(M) = \{$all binary strings containing substring 001$\}$

  $L(M) = \{$all binary strings without substring 001$\}$

- Definitions

  - A language L is regular if there is a DFA M that accepts it

Non-deterministic finite automata's

- Overview

  - If one branch is accepted, that string is accepted

  - i.e if there is a computation of the NFA that accepts it

- Lambda transitions

  - You jump from q_1 to q_2 with the empty string

- Equivalence of Machines

  - If a NFA has k states, the equivalent DFA has 2^k states